

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

MATHEMATICS :

M.1. Quadratic formula.

The roots of the quadratic equation $ax^2 + bx + c = 0$ are.

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

M.2 TRIGONOMETRY

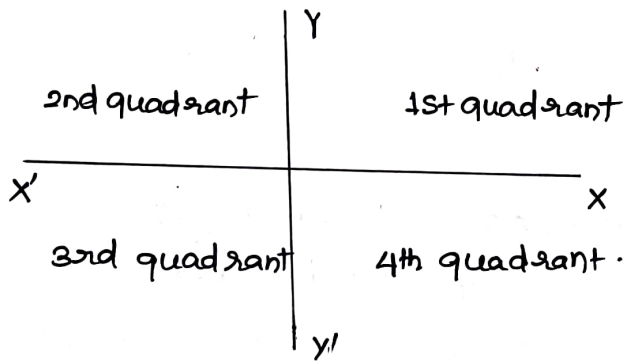
1. - CONVERSION OF ONE SYSTEM INTO ANOTHER :

$$\pi \text{ radians} = 180^\circ$$

$$1 \text{ radian} = \frac{180^\circ}{\pi} = 57.16^\circ$$

$$1^\circ = \frac{\pi}{180^\circ} \text{ radians} = 0.01746 \text{ radians}$$

2. - QUADRANTS



$$3. \rightarrow \sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

$$\sec x = \frac{1}{\cos x}$$

$$\operatorname{cosec} x = \frac{1}{\sin x}$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

4. SIGN OF TRIGONOMETRIC RATIOS IN DIFFERENT QUADRANTS

1st quadrant ($0^\circ - 90^\circ$) = All are +ve

2nd quadrant ($90^\circ - 180^\circ$) = only sine & cosecant +ve

3rd quadrant ($180^\circ - 270^\circ$) = only tangent and cot are +ve.

4th quadrant ($270^\circ - 360^\circ$) = only cosine & secant are +ve.

5. TRIGONOMETRIC RATIOS OF STANDARD ANGLES:

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

6. Allied Angles:

If the sum or difference of two angles is a multiple of 90° , then the two angles are allied angles.

eg. $\theta, -\theta, 90-\theta, 90+\theta, 180-\theta, 180+\theta$ etc are allied angles.

7. RELATION BETWEEN TRIG. RATIOS OF ALLIED ANGLES

1. $\sin(-\theta) = -\sin \theta$

$\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$

$\cos(-\theta) = \cos \theta$

$\sec(-\theta) = \sec \theta$

$\tan(-\theta) = -\tan \theta$

$\cot(-\theta) = -\cot(\theta)$.

2. In case of $(90 \pm \theta)$ and $(270 \pm \theta)$ ratio changes from $\sin$ to $\cos$, $\cos \theta$ to $\sin$, $\tan$ to $\cot$, $\cot$ to $\tan$, $\sec$ to cosec , cosec to $\sec$.

$$\sin(90^\circ + \theta) = \cos \theta$$

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ + \theta) = -\sin \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\tan(90^\circ + \theta) = +\cot \theta$$

$$\cot(90^\circ - \theta) = \tan \theta$$

$$\sin(270^\circ + \theta) = -\cos \theta$$

$$\sin(270^\circ - \theta) = -\cos \theta$$

3. In case of $(180^\circ \pm \theta)$ and $(360^\circ \pm \theta)$ trigonometric ratios remains unchanged

$$\sin(180^\circ \pm \theta) = \mp \sin \theta$$

$$\sin(360^\circ \pm \theta) = \pm \sin \theta$$

$$\cos(180^\circ \pm \theta) = -\cos \theta$$

$$\cos(360^\circ \pm \theta) = \cos \theta$$

$$\tan(180^\circ \pm \theta) = \pm \tan \theta$$

$$\tan(360^\circ \pm \theta) = \pm \tan \theta$$

8. SOME FUNDAMENTAL TRIGONOMETRIC RELATIONS:

$$\rightarrow \sin^2 \theta + \cos^2 \theta = 1 \quad ; \quad \cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

$$\rightarrow \sec^2 \theta - \tan^2 \theta = 1$$

$$\rightarrow \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

9. REDUCTION TO THE FIRST QUADRANT:

For $(\alpha < 90^\circ)$ the following relationships apply :-

REDUCTION TO THE FIRST QUADRANT
For $\alpha < 90^\circ$ the following relationships apply:

Angle Function	$\pm \alpha$	$90^\circ \pm \alpha$	$180^\circ \pm \alpha$	$270^\circ \pm \alpha$	$360^\circ \pm \alpha$
sin	$\pm \sin \alpha$	$\pm \cos \alpha$	$\mp \sin \alpha$	$-\cos \alpha$	$\pm \sin \alpha$
cos	$\pm \cos \alpha$	$\pm \sin \alpha$	$-\cos \alpha$	$\pm \sin \alpha$	$\pm \cos \alpha$
tan	$\pm \tan \alpha$	$\pm \cot \alpha$	$\pm \tan \alpha$	$\pm \cot \alpha$	$\pm \tan \alpha$
cot	$\pm \cot \alpha$	$\pm \tan \alpha$	$\pm \cot \alpha$	$\pm \tan \alpha$	$\pm \cot \alpha$
sec	$\pm \sec \alpha$	$\pm \operatorname{cosec} \alpha$	$-\sec \alpha$	$\pm \operatorname{cosec} \alpha$	$\pm \sec \alpha$
cosec	$\pm \operatorname{cosec} \alpha$	$\pm \sec \alpha$	$\pm \operatorname{cosec} \alpha$	$-\sec \alpha$	$\pm \operatorname{cosec} \alpha$

10. TRIGONOMETRIC FORMULA : (SUM AND DIFFERENCE FOR)

$$(a) \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$(b) \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$(c) \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$(d) \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$(e) \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$(f) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

11. FORMULA FOR MULTIPLE AND SUB MULTIPLE ANGLES

$$(a) \sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$(b) \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta$$
$$= 2 \cos^2 \theta - 1 = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$(c) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$(d) \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

$$(e) \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$(f) \tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

12. FORMULA FOR BISECTION:

Trigonometric ratio of angles in form of $A/2$ in terms of $\cos A$

$$(a) \sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$$

$$(b) \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$$

$$(c) \tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$$

13. TRANSFORMATION OF SUM AND PRODUCTS.

$$(1) 2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$(2) 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$(3) 2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$(4) * 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$(5) \sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B$$

$$(6) \cos(A+B) \cos(A-B) = \cos^2 A - \sin^2 B.$$

14. PRODUCT FORMULA.

$$(a) \sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$$

$$(b) \sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$$

$$(c) \cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$$

$$(d) \cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$$

15. POWERS OF TRIGONOMETRIC FUNCTION.

$$(1) \sin^2 \alpha = \frac{(1 - \cos 2\alpha)}{2}$$

$$(2) \cos^2 \alpha = \frac{(1 + \cos 2\alpha)}{2}$$

$$(3) \sin^3 \alpha = \frac{(3 \sin \alpha - \sin 3\alpha)}{4}$$

$$(4) \cos^3 \alpha = \frac{(3 \cos \alpha + \cos 3\alpha)}{4}$$

$$(5) \sin^4 \alpha = \frac{(\cos 4\alpha - 4 \cos 2\alpha + 3)}{8}$$

$$(6) \cos^4 \alpha = \frac{(\cos 4\alpha + 4 \cos 2\alpha + 3)}{8}$$

$$(7) \sin^5 \alpha = \frac{(10 \sin \alpha - 5 \sin 3\alpha + \sin 5\alpha)}{16}$$

$$(8) \cos^5 \alpha = \frac{(10 \cos \alpha + 5 \cos 3\alpha + \cos 5\alpha)}{16}.$$

16. GENERAL EXPRESSION

(1) If $\sin \theta = 0$ then $\theta = n\pi$

(2) If $\cos \theta = 0$ then $\theta = (2n+1)\pi/2$

(3) If $\tan \theta = 0$ then $\theta = n\pi$

(4) If $\sin \theta = \sin \alpha$ then $\theta = n\pi + (-1)^n \alpha$

(5) If $\cos \theta = \cos \alpha$ then $\theta = 2n\pi \pm \alpha$

(6) If $\cos \theta = \cos \alpha$ then $\theta = n\pi + \alpha$

where n is any integer.

Quadratic equation: $\sin^2 \theta = \sin^2 \alpha$, $\cos^2 \theta = \cos^2 \alpha$ & $\tan^2 \theta = \tan^2 \alpha$

then $\theta = n\pi \pm \alpha$

17. INVERSE TRIG:

If $\sin y = x$ then $y = \sin^{-1} x$

(1) $\sin^{-1}(\sin \theta) = \theta$

(2) $\cos^{-1}(\cos \theta) = \theta$

(3) $\tan^{-1}(\tan \theta) = \theta$

(4) $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$; $\cot^{-1} x = \tan^{-1} \frac{1}{x}$; $\sec^{-1} x = \cos^{-1} \frac{1}{x}$

(5) $\sin^{-1}(-x) = -\sin^{-1} x$

(6) $\cos^{-1}(-x) = \pi - \cos^{-1} x$

(7) $\tan^{-1}(-x) = -\tan^{-1} x$

(8) $\sin^{-1} x + \cos^{-1} x = \pi/2$

(9) $\tan^{-1} x + \cot^{-1} x = \pi/2$

(10) $\operatorname{cosec}^{-1} x + \sec^{-1} x = \pi/2$

STANDARD DERIVATIVES

$$(1) \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$(2) \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$(3) \cdot \frac{du}{dx} = \frac{du}{dy} \frac{dy}{dx}$$

$$(4) \cdot \frac{d}{dx}(ax+b)^n = n \cdot (ax+b)^{n-1} \cdot a$$

$$(5) \frac{d}{dx}(e^x) = e^x$$

$$(*) (6) \cdot \frac{d}{dx}(a^x) = a^x \log_e a.$$

$$(7) \frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$(*) (8) \frac{d}{dx}(\log_a x) = \frac{1}{x} \log_e a = \frac{1}{x \log a}$$

$$(9) \frac{d}{dx}(\sin x) = \cos x$$

$$(10) \frac{d}{dx}(\cos x) = -\sin x$$

$$(11) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(12) \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$(13) \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$(14) \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x.$$

$$P (15) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$P (16) \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$P (17) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(18) \frac{d}{dx} (\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(19) \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$(20) \frac{d}{dx} (\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

$$(21) \frac{d}{dx} (\sinh x) = \cosh x$$

$$(22) \frac{d}{dx} (\cosh x) = \sinh x.$$

→ STANDARD INTEGRALS:

$$(1) \int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$(2) \int k dx = kx + c$$

$$(3) \int \frac{dx}{x} = \log x + c$$

$$(4) \int \sin x dx = -\cos x$$

$$(5) \int \cos x dx = \sin x$$

$$(6) \int \tan x dx = -\log \cos x \quad (\text{or}) \log \sec x$$

$$(7) \int \cot x dx = \log \sin x$$

$$(8) \int \sec x dx = \log (\sec x + \tan x)$$

$$(9) \int \operatorname{cosec} x dx = \log (\operatorname{cosec} x - \cot x)$$

$$(10) \int \sec^2 x dx = \tan x$$

$$(11) \int \operatorname{cosec}^2 x dx = -\cot x$$

$$(12) \int a^x dx = \frac{a^x}{\log a}$$

$$(13) \int (ax+b)^n x dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c ; n \neq -1$$

$$(14) \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + c$$

$$(15) \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + c$$

$$(16) \int \sec x \cdot \tan x dx = \sec x + c$$

$$(17) \int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + c$$

$$(18) \int \frac{dx}{ax+b} = \frac{1}{a} \log(ax+b)$$

$$(19) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c$$

SOME OTHER FORMS:

$$\textcircled{1} (20) \int \frac{dx}{\sqrt{1-x^2}} dx = \sin^{-1} x + c = -\cos^{-1} x + c$$

$$\textcircled{1} (21) \int \frac{dx}{1+x^2} = \tan^{-1} x + c = -\cot^{-1} x + c$$

$$\textcircled{1} (22) \int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c = -\operatorname{cosec}^{-1} x + c$$

SOME MORE STANDARD FORMS:-

$$\textcircled{1} 1. \int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$$

$$\textcircled{1} 2. \int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$$

$$\textcircled{1} 3. \int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) .$$

4.

STANDARD WAVEFORMS

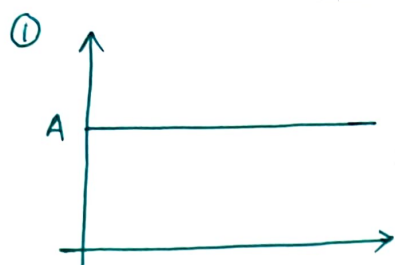
V_{avg}

V_{rms}

Form factor

Peak factor

V_{pp}



A

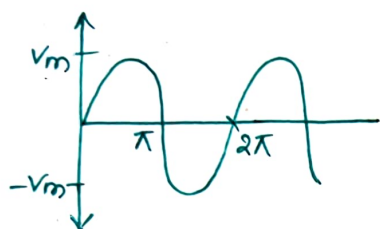
A

1

1

A

(2) $V_m \sin(\omega t + \theta)$



(full cycle)

0

(half cycle)

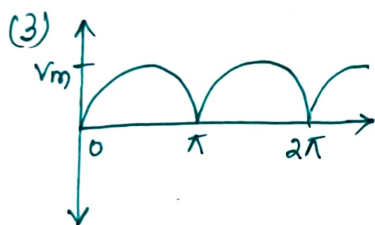
$$\frac{2V_m}{\pi}$$

$$\frac{V_m}{\sqrt{2}}$$

$$\frac{\pi}{2\sqrt{2}} = 1.11$$

$$\sqrt{2} = 1.41$$

$$2V_m$$



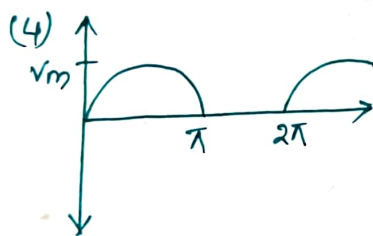
$$\frac{2V_m}{\pi}$$

$$\frac{V_m}{\sqrt{2}}$$

$$1.11$$

$$\sqrt{2}$$

$$V_m$$



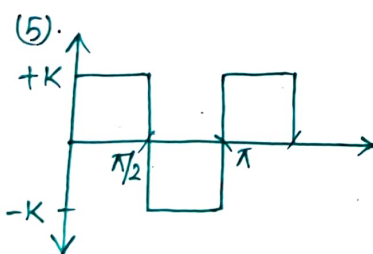
$$\frac{I_m}{\pi} \text{ or } \frac{V_m}{\pi}$$

$$\frac{V_m}{2}$$

$$1.57$$

$$2$$

$$V_m$$



(full cycle)

0

(half cycle)

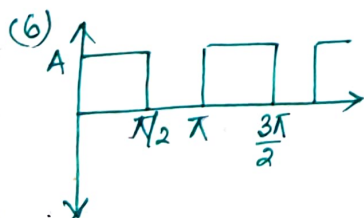
K

K

1

1

2K



$$\frac{A}{2}$$

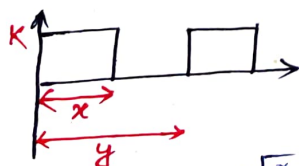
$$\frac{A}{\sqrt{2}}$$

$$\sqrt{2}$$

$$\sqrt{2}$$

$$A$$

NOTE:



$$AVG = K\left(\frac{x}{y}\right) \quad RMS = K\sqrt{\frac{x}{y}}$$

STANDARD WAVEFORM

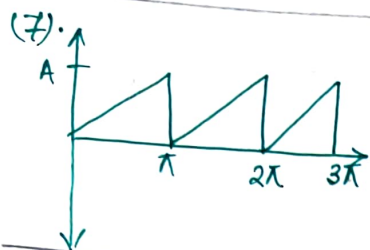
Vavg

Vrms

FF
(RMS/AVG)

PF
(MAX/RMS)

Vpp



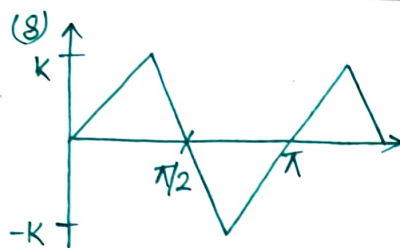
$$\frac{A}{2}$$

$$\frac{A}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$A$$



(Half cycle)

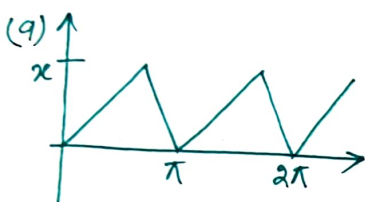
$$\frac{K}{2}$$

$$\frac{K}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$2K$$



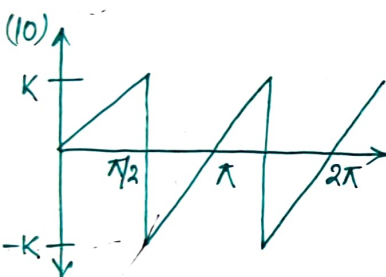
$$\frac{x}{2}$$

$$\frac{x}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$x$$



(Full cycle)

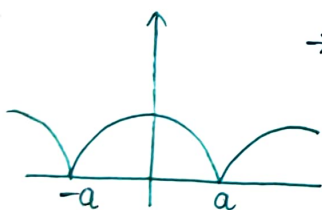
$$\frac{K}{2}$$

$$\frac{K}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}}$$

$$\sqrt{3}$$

$$2K$$



$$\rightarrow V_{avg} = \frac{\pi a^2}{2 \times 2a}$$

$$\rightarrow V_{avg} = \frac{\pi a}{4}$$

$$\rightarrow V_{rms} = \sqrt{\frac{2}{3}} a$$

SN	PARAMETER	HALF (RECTIFIER)	FULL	BRIDGE
1	No of diodes	1	2	4
2	PIV	V_m	$2V_m$	V_m
3	I_{RMS}	$I_m/2$	$I_m/\sqrt{2}$	$\frac{I_m}{\sqrt{2}}$
4	$I(AVG)$	I_m/π	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
5	DC O/P V/G	$\frac{V_m}{\pi}$	$\frac{2V_m}{\pi}$	$\frac{2V_m}{\pi}$
6	Ripple factor RF: $\sqrt{FF^2 - 1}$	1.21	0.482	0.482
7	Ripple freq	f	2f	2f
8	FORM FACTOR	1.57	1.11	1.11
9	Rectification efficiency(%)	40.6%	81.2%	81.2%
10	TUF	0.287	0.693	0.812

$$1e^- = 1.602 \times 10^{-19} C; \quad 1C = \frac{1}{1.602 \times 10^{-19}} \text{ no of elec} = 6.24 \times 10^{18} \text{ elec}$$

$$6.24 \times 10^{18} \text{ electrons} = 1C$$

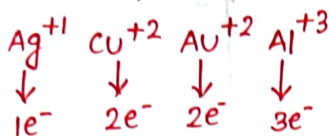
$$1 \text{ elec} = \frac{1}{6.24 \times 10^{18}} C$$

(NETWORKS)

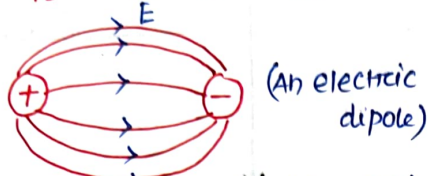
NETWORK THEORY

BASIC CONCEPTS OF NETWORKS:

→ The four best conductors are:



→



The force exerted by field on each free electron is given by $F = E \cdot e$ (N)
where $e = -1.6 \times 10^{-19} C$

The amount of charge flowing is given by
 $q = ne$ (C) i.e. $q(t) = n(t) \cdot e$ (C).

$$\rightarrow i = \frac{dq}{dt} \text{ (A)}$$

$$\rightarrow j = I/A \text{ (A/m}^2\text{)} \text{ where } j = \text{current density}$$

$A = \text{Area.}$

$$\rightarrow \vec{F} = \vec{E} \cdot e \text{ (N)}$$

$$\rightarrow q = ne \text{ (C)}$$

$$\rightarrow i = \frac{dq}{dt} \text{ (A)}$$

$$\rightarrow j = I/A \text{ (A/m}^2\text{)}$$

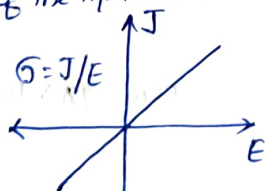
OBSERVATION

$$E \uparrow \Rightarrow F \uparrow \Rightarrow n \uparrow \Rightarrow q \uparrow \Rightarrow i \rightarrow j \uparrow \text{ (linearly).}$$

AS $E=0$ all other parameters are zero
ie all the electrons are at rest.

$$\text{OHMS LAW: } \boxed{j \propto E} \quad \vec{j} = \sigma \vec{E}$$

σ - conductivity of the materials.



Amount of power dissipated within a conductor: $P = j \cdot E$ put $j = \sigma E$

$$\boxed{P = \sigma E^2} \text{ W/m}^3$$

Getting started:

CIRCUIT: Loop or Mesh ie (closed path)

Interconnection of ckt is a Network.

eg - Network is like a building
where our ckt are like rooms in that building.

(a) Resistance (R)

Since $j = \sigma E$

$$\frac{I}{A} = \sigma \frac{V}{l}$$

where $E = (\text{voltage per unit length})$

$$V = \frac{I \times l}{\sigma A} \text{ or } I \times \frac{l}{\sigma A}$$

$$\boxed{V = RI}$$

$$\text{where } R = \frac{l}{\sigma A} \text{ (}\Omega\text{)}$$

As $t \uparrow, l \uparrow$ (length), $A \uparrow, \sigma \downarrow$, So $R \uparrow$

$$\rightarrow R_t = R_0(1 + \alpha t)$$

where α - temperature coefficient in $^{\circ}C$

$\uparrow \uparrow R \uparrow$ for all conductors.

$$\rightarrow \boxed{i = GV} \text{ where } G = 1/R \text{ - (3rd form Ohms law)}$$

$$\rightarrow \boxed{V = R \cdot \frac{dq}{dt}} \rightarrow \text{(4th form Ohms law)}$$

→ Electric power (P)

$$P = \frac{dW}{dt} = \frac{dW}{dq} \cdot \frac{dq}{dt} = V \cdot I \text{ (Watts)}$$

$$\text{where } \frac{dW}{dq} = \text{Voltage}$$

$$\frac{dq}{dt} = \text{current.}$$

→ Electrical energy (W)

$$\frac{dW}{dt} = P$$

$$dW = P dt$$

$$\boxed{W = \int P dt \text{ (J)}}$$

$$\text{FOR } R: P = VI = I^2 R = \frac{V^2}{R} \text{ (Watts)}$$

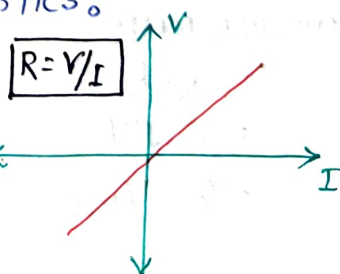
$$W = \int P dt = \int i^2 R dt = \int \frac{V^2}{R} dt = \int V I dt \text{ (J)}$$

→ V-I CHARACTERISTICS:

→ Linear

→ passive

→ time invariant



(B) INDUCTOR:

Total flux produced

$$\Psi = N\Phi \text{ (Webers)}$$

Where N - No of turns

Φ - flux per turn

The total flux produced is proportional to the current flowing through the coil

$$\Psi \propto i$$

$$\Psi = Li \leftarrow \text{(Ohms law)}$$

Where L Inductance parameter of coil.

$$\rightarrow V = L \frac{di}{dt} \rightarrow I = \frac{1}{L} \int_{-\infty}^t v dt$$

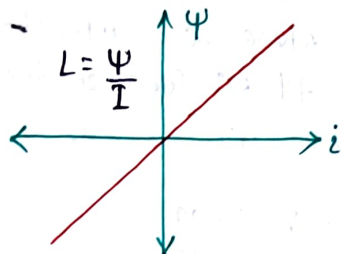
$$\rightarrow \text{Energy stored} = \frac{1}{2} LI^2$$

$$\rightarrow \text{Maxm energy stored} = \frac{1}{2} LI_{\text{max}}^2$$

→ Linear

→ Passive

→ Bilateral

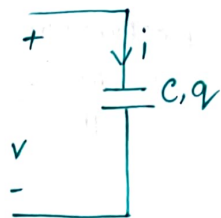


(C) CAPACITOR:

$$I = \frac{dq}{dt}$$

$$q \propto V$$

$$q = CV$$



$$\rightarrow I = \frac{cdv}{dt} \rightarrow v = \frac{1}{C} \int_{-\infty}^t i dt$$

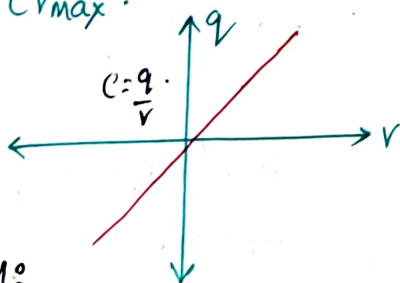
$$\rightarrow W = \frac{1}{2} CV^2$$

$$\rightarrow W_{\text{max}} = \frac{1}{2} CV_{\text{max}}^2$$

→ Linear

→ passive

→ Bilateral



ANOTHER FORM:

$$Z = Z_R = R$$

$$= Z_L = j\omega L$$

$$= Z_C = 1/j\omega C$$

(*) Average power can also be calculated as

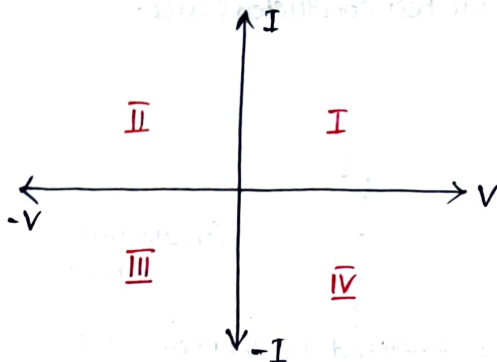
$$\text{(for waveforms)} P_{\text{avg}} = \frac{\text{The energy absorbed over 1 period}}{\text{Time period}}$$

→ Bilateral components:

R, L, C , transmission lines, communication lines

→ UNILATERAL components:

→ Diodes, thyristors, MOSFET



→ $I = \text{III}$ Symmetrical → Bilateral

→ $I \neq \text{III}$ Asymmetrical → Unilateral

→ for any network or ckt $m = b - n + 1$

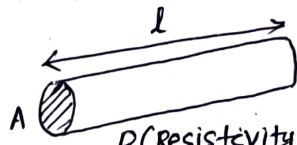
mesh = branch - node + 1

→ The minimum no of equations req to solve a n/w is $b - n + 1$

→ Meshes are independent loops.

→ RESISTANCE

$$R = \frac{\rho l}{A}$$



ρ (resistivity) ($\Omega\text{-m}$)

ρ depends upon the Nature of the Material

$$R_t = R_0 (1 + \alpha t)$$

Where α temp coefficient of material

α = +ve for conductors

α = -ve for semiconductors.

→ INDUCTANCE

$$L = \frac{\mu N^2 A}{l}$$

$$\mu = \mu_0 \mu_r$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

→ CAPACITANCE

$$C = \frac{\epsilon A}{d}$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$